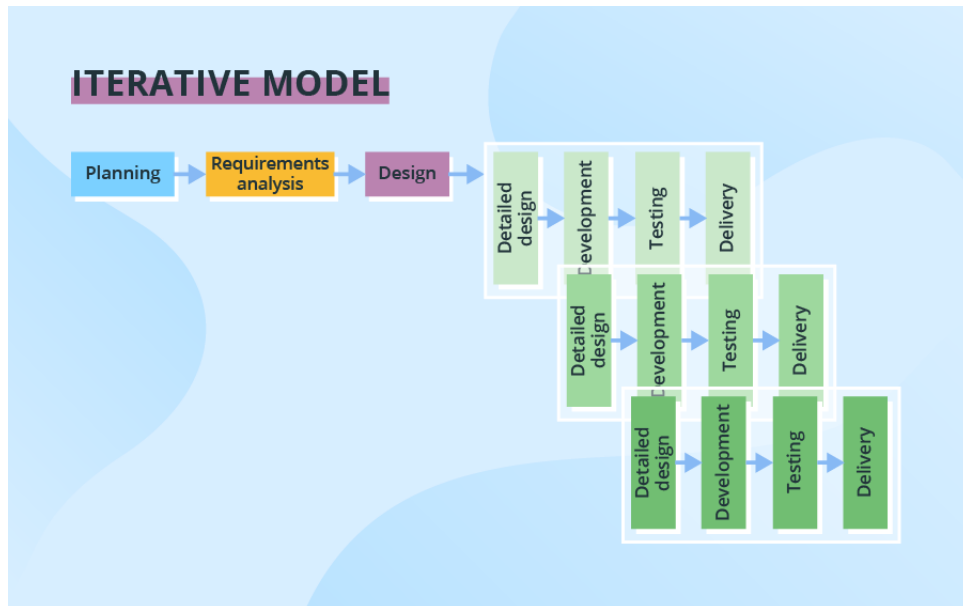


Iterative Model

Continuous Refinement for Evolving Needs



The **Iterative Model** emphasizes continuous improvement through repeated development cycles. Each iteration produces a refined version of the software by incorporating feedback and lessons from previous iterations.

Key Characteristics

- Development is divided into multiple iterations or cycles.
- Each iteration includes requirements, design, coding, and testing.
- Later iterations refine and expand earlier functionality.
- Feedback from each cycle guides future improvements.

Benefits

- Provides flexibility to accommodate evolving requirements.
- Enables early issue detection and correction.
- Uses client feedback to refine design and improve usability.

Challenges

- May lead to scope creep if project limits are not clearly managed.
- Requires active and continuous client involvement.
- Can consume more time if iterations are not well structured.

Best Suited For

- Projects with dynamic or frequently changing requirements.
- Software needing continuous user feedback and improvement.

Not Ideal For

- Projects that demand complete, fixed requirements before starting.
- Systems where the full specification must be defined upfront.

Summary

The Iterative Model allows software to evolve through cycles of feedback, testing, and refinement. It promotes adaptability and quality but requires clear control of scope and strong stakeholder collaboration.